

ICS4U1 CPT Common Requirements

Below are some features that all CPT projects must have:

- Panel size should be exactly 1280x720 pixels. The MenuBar and Frame can go around.
- All programs must have a chat area during gameplay. You can have a JTextField and JTextArea for chat OR you can have the player press the enter key, type their message, press enter to send the message out (and the receiving messages can be overlaid over the gameplay).
- All programs must have one or more data files that contain the data for the objects of the game. The data files can be in csv or txt format depending on the type and/or quantity of objects. Ex: Maybe you are creating a fighting game like street fighter, the data file called fighters.csv might have a line “hero name, health amount, damage amount, special move” → “cole, 1, 1, laugh”, “cadawas, 1000, 9001, mock cole”
- All programs must have a network component. As a result, all programs must have a network “language” that distinguishes between chat messages, game messages, movement, attack, etc.
- All programs must use an animated panel in their program that refreshes at 60fps.
- All programs must also use multiple JComponents (Buttons, Text Fields, etc)
- One window only. You can swap out the panels if you need, but ensure that there is one main game frame. Only exception: You can open a JFileChooser
- All programs should try to follow the Model View Controller paradigm of programming as best as possible. The program should therefore should have:
 - A “Model” java class where the main data of the game is stored
 - A “View” java class where the user interface (buttons, frames, panels) are defined
 - A “Controller” that connects the Model and View together
 - In your programs, you will most likely have a model, but a combined view/controller
- All images need to be original. These programs might be published on a public website so let’s avoid copyright issues. AI generated images are ok.
- Only 1 JFrame allowed

Client Requirements:

Core Mechanics/Game Logic(See Figure 1.1/1.2 for Flowcharts):

- Grid System: Two 10x10 Grid(Left Grid is for Placing Ships, Right Grid is for shooting enemy ships)
- Fleet Composition - (*Ships must be drawn out, and not simple oval shapes.*)

- Carrier (5 holes)
- Battleship (4 holes)
- Cruiser (3 holes)
- Submarine (3 holes)
- Destroyer (2 holes)



- How to Play:
 - Players connect to servers and place ships on their side of the grid.
 - Players alternate shooting missiles to the opponents ships.
 - Whoever sinks the opponent’s ship first wins!

Feature Specifications(See Figure 2 for Layout):

- Preparation Phase:
 - Drag and Drop ships onto the ocean, also able to rotate the ships(r).
- Combat Phase:
 - Click the coordinate and not type out the coordinate when attacking enemy ships.
- Real-Time Chat - Side Panel Chat that allows players to communicate.

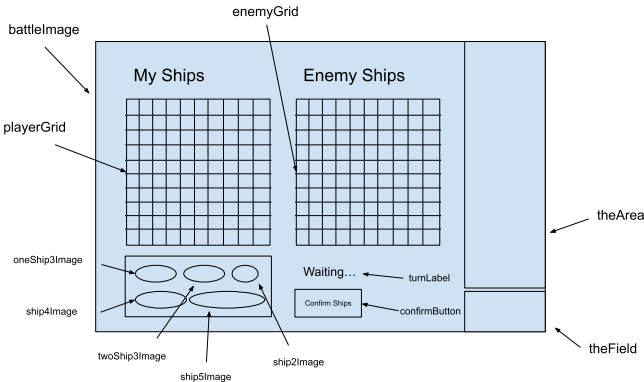
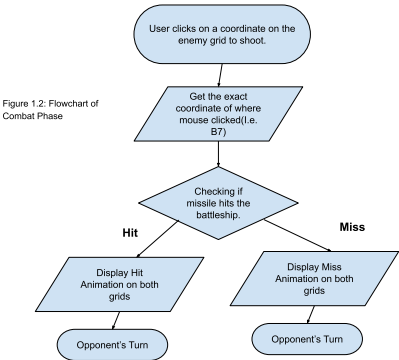
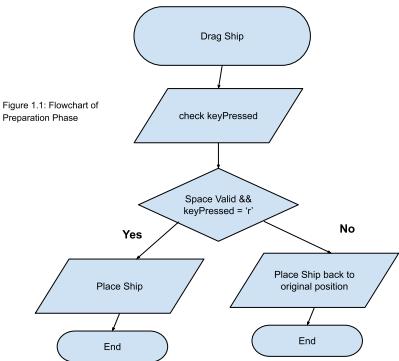


Figure 2: The user drags and drops ships on the “My Ships” grid, then presses Confirm Ships. Players takes turns pressing grid squares on the “Enemy Ships” grid, and are able to send and receive messages through the text field and area